Genuine Multipartite Entanglement Analysis

```
Model                              Ground-state properties           GME summary
-----                              ----------------------           -----------
Hubbard ring (1-D, PBC)            E0                 : -4.60352630  Extracted n-spin RDM : n = 4
                                   E1                 : -4.29999276
System parameters                                                    Purity:
-----------------                  <S2>               : -0.000000      gamma = Tr(rho^2)
Sites (L)          : 8             S                  : -0.0000
Electrons (Ne)     : 8             <Sz>               : 0.000000    Separable cut criterion:
Hopping (t)        : 1.0           Multiplicity 2S+1  : 1.0           gamma_A = 1  <->  rho = rho_A x rho_B
Interaction (U)    : 4.0           Spin state         : Singlet (S=0)
Mode               : uniform                                       C_GME (Troiani et al.):
n-body order       : 4             Double occupancy                  C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))
                                   ----------------
                                   <D> total          : 0.759406    Convex roof (mixed):
                                   <D> / site         : 0.094926      C_GME(ρ) = min Σ pᵢ C_GME(|ψᵢ⟩)


                                                                     Date: 2026-04-13  20:48
```